

European Cross-Commodity Risk Monitor

Gas Tightness · Carbon Supply Signal · Power Curve Implications

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Market Narrative

EU gas storage sits at 36.3% full, a staggering 19.2 percentage points below the five-year seasonal average for this time of year, and the arithmetic is unforgiving: hitting the 90% November target requires injecting 3,704 GWh/day against a current pace of just 2,100 — a shortfall of 1,604 GWh/day that the market has shown no sign of closing. TTF at €50.03/MWh, up 40% year-on-year and sitting in the upper half of its 52-week range, is already pricing a degree of stress, but the backwardated curve structure is actively working against storage operators by making summer injection uneconomic relative to winter delivery. The base case is that Europe enters winter 2026/27 materially undersupplied. Carbon is holding at €75.02/t, contributing €29.56/MWh to the variable cost of every megawatt-hour produced by a gas-fired turbine — roughly a third of the current Cal+1 power price. The more telling signal is what Cal+1 German power is implying about carbon: back-solving the clean spark spread equation gives an implied EUA of just €44.30/t, a €30.72 discount to where carbon actually trades. That gap means forward power is either pricing a collapse in EUA that the supply fundamentals don't support, or assuming enough renewable displacement to make carbon largely irrelevant in the 2027 dispatch stack. Both look optimistic from here, particularly with the MSR mechanically tightening auction supply from September and the July ETS Directive review carrying hawkish tail risk. German Cal+1 at €92.75/MWh looks cheap relative to the fundamental stack. The negative CSS of –€12.12/MWh tells you the forward curve is not being driven by gas plant hedging — generators aren't locking in output at these levels, which means the price discovery is thin and the curve is vulnerable to a sharp repricing the moment winter risk becomes front-page news. A cold October with storage at 70% rather than 90% would close the implied EUA gap and add €15–20/MWh to Cal+1 in short order. The one risk that changes the call is a Hormuz resolution: normalised LNG shipping would compress TTF by €8–12/MWh and Cal+1 follows 1:1, unwinding the entire thesis.

1 | Gas Tightness

TTF M+1	€49.30/MWh (+35.6% YoY)
EU Storage	36.3% (+6.2pp vs 5-yr avg)
Required injection	3704 GWh/day Current: 2100 GWh/day Gap: +1604
Norwegian supply	Plateau ~114.9 Bcm 2025; 2026 guide 110–120 Bcm
LNG shock	Qatar force majeure ~17% of global liquefaction (March 2026)

Chart 1 — EU Natural Gas Storage vs Seasonal Norms | Source: GIE AGSI+



Chart 1: EU gas storage fill % vs 5-year seasonal band. Current: 36.3% (+6.2pp vs avg). Source: GIE AGSI+.

2 | Carbon Signal

EUA front-year	€64.76/t
Carbon cost per MWh	€25.52/MWh (EUA × 0.394 tCO ₂ /MWh)
MSR withdrawal	275.5 Mt removed Sep 2025–Aug 2026 Supply tightens Sep 2026
ETS Directive review	July 2026 Key policy catalyst

3 | Power Curve Implications

Clean Spark Spread:

$CSS = Power - (Gas \div \eta) - (EF \times Carbon) = 104.60 - 100.35 - 25.52 = \text{€-21.26/MWh}$

Implied EUA (solving CSS = 0):

$EUA^* = (Power - Gas/\eta) \div EF = \text{€10.80/t}$ vs spot €64.76/t → gap: €53.96/t

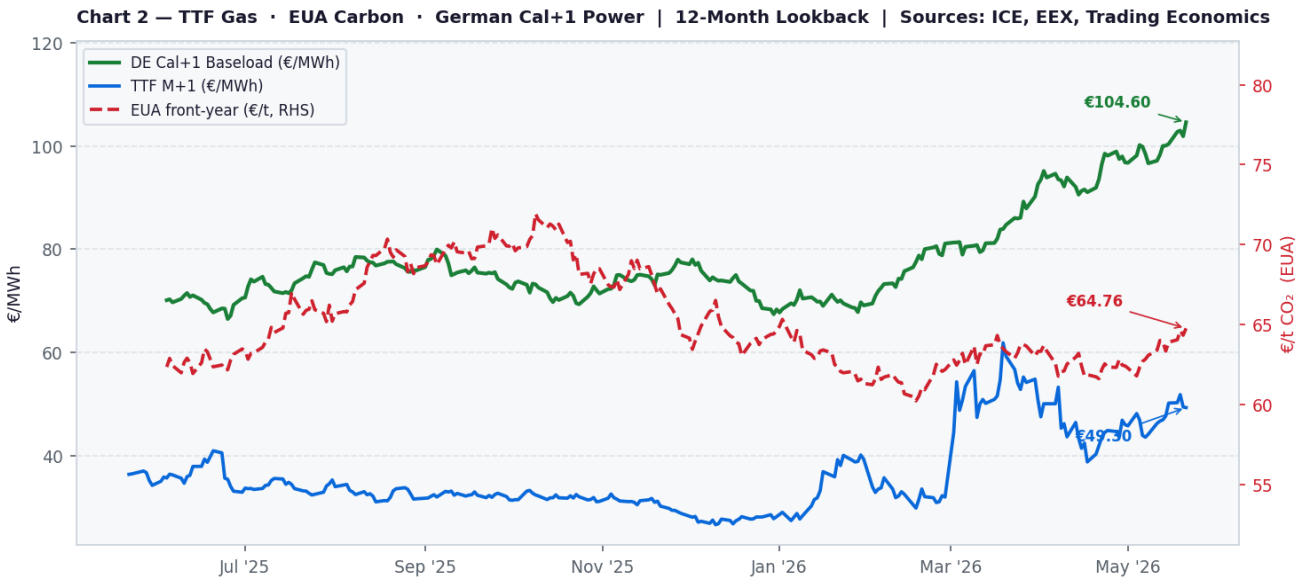


Chart 2: TTF M+1 (blue), EUA front-year (red, RHS), German Cal+1 baseload (green). 12-month lookback. Sources: ICE, EEX, Trading Economics.

4 | Monitor Metrics Dashboard

Metric	Value	Signal	Relevance
TTF M+1 (€/MWh)	€49.30 (+35.6% YoY)	▲ BULLISH	Primary power price input
EU Storage vs 5-yr avg	36.3% (+6.2pp)	▼ BEARISH	Low buffer amplifies demand shocks
Required injection pace	3704 GWh/d	▼ BEARISH	Gap vs target = winter risk
EUA front-year (€/t)	€64.76	→ NEUTRAL	+€30/MWh carbon cost in power
Carbon cost per MWh	€25.52/MWh	→ NEUTRAL	Direct variable cost floor
Clean Spark Spread	€-21.26/MWh	▼ BEARISH	Gas plant profitability
German Cal+1 baseload	€104.60/MWh	▲ BULLISH	Forward power curve anchor

Metric	Value	Signal	Relevance
Implied EUA in Cal+1	€10.80/t (gap: €53.96)	▲ BULLISH	Carbon unpriced in forward power

5 | Risk Skew

Upside catalysts:

▲ TTF at €49.30/MWh — any Hormuz re-escalation or Norwegian outage could push toward 52-week high of €62/MWh.

Downside catalysts:

- ▼ CSS at €-21.26/MWh — warm autumn + unhedged generators = spot volatility risk.
- ▼ Hormuz resolution / US-Iran deal — TTF -€8–12/MWh; Cal+1 power follows 1:1.
- ▼ US LNG wave on schedule — 2027 supply easing weighs on Cal+2 and beyond.